

LANGUAGE: C++ PROGRAMMING LEVEL: BASIC-INTERMEDIATE

Day	Topic	Subtopics	Interview Coding Problems (LeetCode / HackerRank)	End of Module Use Case
Day 1	C++ Fundamentals	Introduction to C++, Program Structure, Data Types, Variables, Operators, Input/Output, Type Casting	LeetCode: Palindrome Number, Fizz Buzz HackerRank: Basic Data Types, Input and Output	Student Grade Calculator
Day 2	Control Statements & Loops	if-else, switch-case, for, while, do-while, Nested Loops, break & continue	LeetCode: Valid Perfect Square, Power of Two HackerRank: Loops, Conditional Statements	ATM Menu Driven System
Day 3	Functions & Arrays	Functions, Function Parameters, Default Arguments, Inline Functions, 1D/2D Arrays	LeetCode: Two Sum, Maximum Subarray HackerRank: Arrays Introduction	Student Marks Analysis System
Day 4	Strings & Pattern Programming	Strings, String Functions, Character Arrays, Pattern Problems, Command Line Arguments	LeetCode: Valid Anagram, Longest Common Prefix HackerRank: StringStream, Strings	Password Validation System
Day 5	Introduction to OOP	Classes & Objects, Access Specifiers, Constructors, Destructor, this Pointer	LeetCode: Design Parking System, Final Value of Variable After Operations HackerRank: Classes and Objects	Library Management System
Day 6	OOP Concepts	Encapsulation, Inheritance, Types of Inheritance, Function Overloading, Operator Overloading	LeetCode: Climbing Stairs, Merge Sorted Array HackerRank: Inherited Code	Employee Payroll System
Day 7	Polymorphism & Abstraction	Method Overriding, Virtual Functions, Runtime Polymorphism, Abstract Classes, Interfaces	LeetCode: Valid Parentheses, Single Number HackerRank: Virtual Functions	Banking Management System
Day 8	Pointers & Dynamic Memory	Pointers, References, Pointer Arithmetic, new/delete, Dynamic Memory Allocation, Memory Leaks	LeetCode: Move Zeroes, Remove Element HackerRank: Pointer Exercises	Dynamic Inventory Management
Day 9	Structures, Files & Exception Handling	Structures, File Handling, Reading/Writing Files, Exception Handling (try-catch-throw)	LeetCode: Contains Duplicate, Rotate Array HackerRank: Basic File I/O	Student Database using Files
Day 10	STL Basics	vector, list, stack, queue, pair, iterators, Algorithms Library (sort, find)	LeetCode: Two Sum, Binary Search HackerRank: STL Vector, STL Queue	Online Shopping Cart System
Day 11	Advanced STL & Problem Solving	map, unordered_map, set, unordered_set, Lambda Functions, Searching & Sorting Problems	LeetCode: Group Anagrams, Majority Element, Top K Frequent Elements	Student Attendance Tracking System

Day	Topic	Subtopics	Interview Coding Problems (LeetCode / HackerRank)	End of Module Use Case
			HackerRank: Maps-STL, Sets-STL	
Day 12	Interview Coding Practice	Time Complexity, Space Complexity, Arrays & String Interview Problems, Hashing, Debugging, Output Prediction	LeetCode: Best Time to Buy & Sell Stock, Merge Intervals, Valid Sudoku HackerRank: Balanced Brackets, Left Rotation, Sparse Arrays	Campus Interview Coding Round Simulation
Day 13	Advanced Coding + RDBMS Concepts	First Half – Coding Practice: Mock Coding Round, Optimization Techniques, STL-Based Questions, Debugging Round. Second Half – RDBMS Concepts: DBMS vs RDBMS, ER Model, Keys, Normalization, SQL Basics, SELECT, INSERT, UPDATE, DELETE, Joins	LeetCode: Rotate Array, Valid Anagram, Contains Duplicate HackerRank: SQL Basic Select, Dynamic Array, Weather Observation Station	Student Database Management System
Day 14	Cloud Basics, GitHub, Docker, Kubernetes & CI/CD	Overview of Cloud Computing, Types of Cloud Services, Git & GitHub Basics, clone/push/pull/merge conflicts, Docker Basics, Docker Images & Containers, Kubernetes Basics, K8S Commands, CI/CD Pipelines, GitHub Actions/Jenkins Basics	LeetCode: Single Number, Move Zeroes, Merge Sorted Array HackerRank: Linux Shell Basics, Git Tutorials, 30 Days of Code	End-to-End DevOps Deployment Pipeline using GitHub + Docker + Kubernetes

Be Agile..Be in Demand